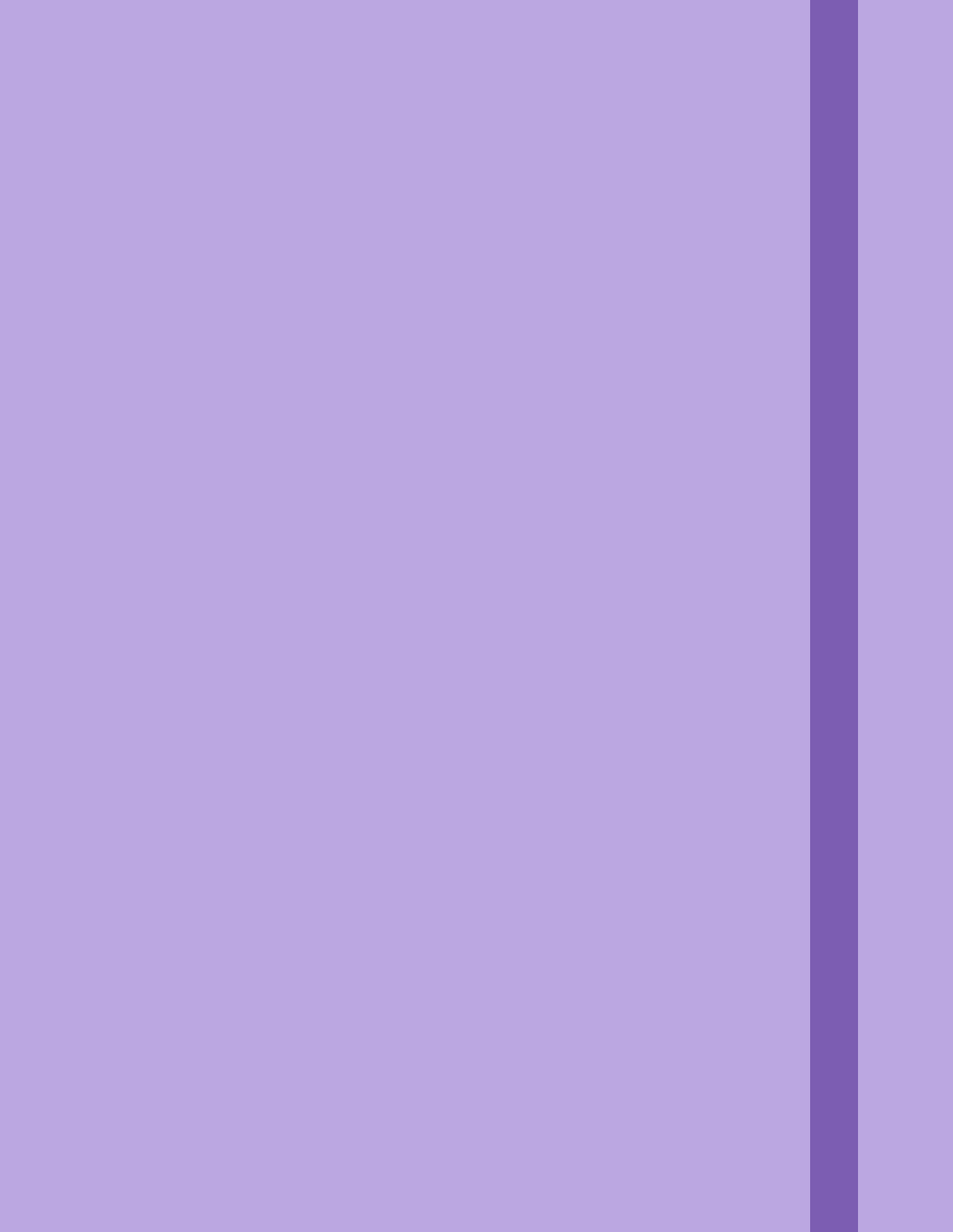




Linear Algebra - Part 0 | Tensors.

Tensor is data structure, container for storing the numbers.

Dimension 0 $\rightarrow$ Scalar

List of no. $\rightarrow$ Vector.

2D no.s $\rightarrow$ Matrix.

General term $\rightarrow$ Tensors.

0D Tensor | Scalar :

Storing a single no. [2], [3]

Scalar $\rightarrow$ 0D Tensor

`a = np.array(4)` # Tensor $\rightarrow$ 0D
`a.ndim` # 0D $\rightarrow$ 0

1D Tensor | Vector :

[1, 2, 3, 4, 5] $\rightarrow$ 1D Tensor / Vector / 1D array or array.

`ndim` $\rightarrow$ 0D = 1.

Axis $\rightarrow$ 1

No. of Axis = Rank = Dimension.

`arr = np.array([1, 2, 3])`
`arr.ndim` 0D $\rightarrow$ 1

3D Tensor | Vector

$\hookrightarrow$ Dimension = 3

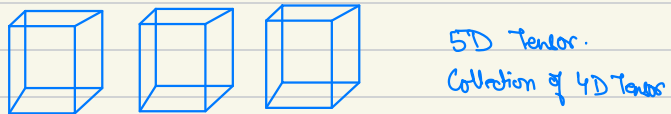
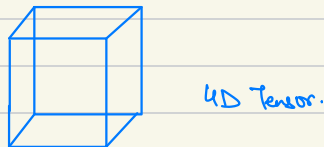
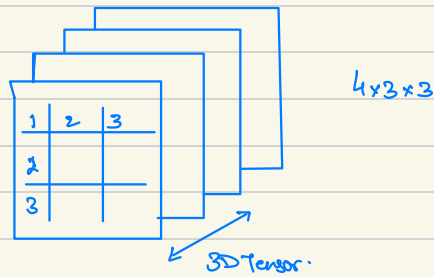
(3 elements in arr)

$[0, 1, 2, 3] \rightarrow \text{Scalars} \xrightarrow{\text{become}} \text{Vectors.}$

$[1, 2, 3] \quad [4, 5, 6] \quad [7, 8, 9]$

$$\begin{bmatrix} [1, 2, 3] \\ [4, 5, 6] \\ [7, 8, 9] \end{bmatrix} \xrightarrow{2D} \text{Rank} = 2.$$

`mat = np.array([ [1, 2, 3], [4, 5, 6], [7, 8, 9] ])`



In ML generally we have 0 to 5D Tensors.

Rank : No. of Dimensions = Axis

Shape (2,2)

Shape (2,3)

Rank = 2

↓
row & col.

Example of 1D Tensor :

Students (10000 students data)

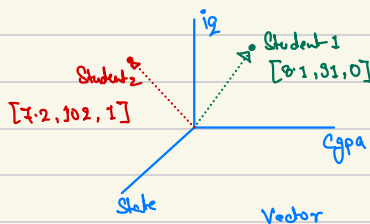
CGPA	ID	State	Placement
8.1	91	WB	1

Let there be only 2 states $\rightarrow$ WB = 0 & KR = 1

1D Tensor / Vector

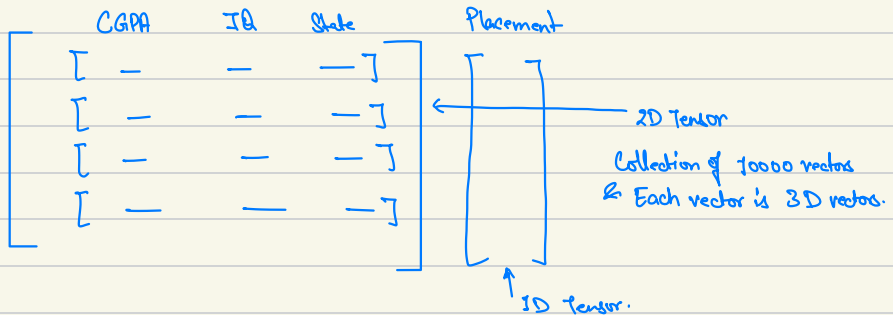
CGPA	ID	State	Placement
8.1	91	0	1

Tensor $\rightarrow [8.1, 91, 0]$ $\xrightarrow{\text{1D Tensor but 3D vector.}}$ Vector = 3D dimension
Axis = 1



Example of 2D Tensor :

Students (10000 students data)



Example of 3D Tensor :
NLP

Hi Nitish
Hi Rahul
Hi Ankit

	Hi	Nitish	Rahul	Ankit
Vector →	1	0	0	0
	0	1	0	0
	0	0	1	0
	0	0	0	1

Hi Nitish.
[[1, 0, 0, 0], [0, 1, 0, 0]] ← 2D Tensor.
Hi Rahul
[[1, 0, 0, 0], [0, 0, 1, 0]]
Hi Ankit
[[1, 0, 0, 0], [0, 0, 0, 1]]

$$\begin{bmatrix} [1, 0, 0, 0], [0, 1, 0, 0], \\ [1, 0, 0, 0], [0, 0, 1, 0], \\ [1, 0, 0, 0], [0, 0, 0, 1] \end{bmatrix}$$

3D Tensor.

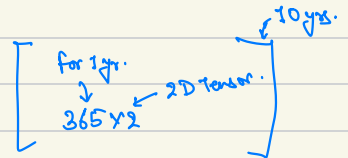
Shape $\rightarrow (3 \times 2 \times 4)$

Time Series Data.

e.g. Share market data.

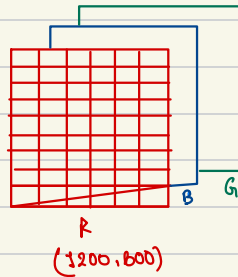
Stocks.

Highest Price Lowest Price



Example of 4D Tensors:

e.g.) Image based data. $\rightarrow$ cv.



$[3, 1200, 800]$

$[50, 3, 1200, 800]$ 4D Tensor.

Example of 5D Tensors:

g) Videos.

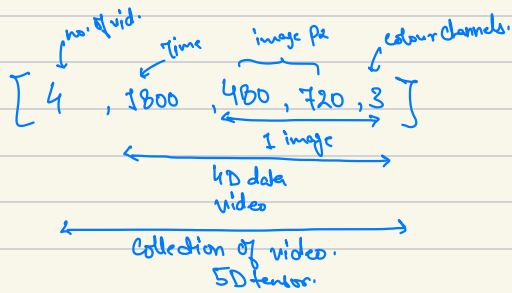
→ Collection of images at high speed.

↳ 4 videos

↳ 60 sec video

↳ 30 fps

↳ 480 px. → 480x720px (3 Channels)



Linear Algebra — Part 1 | Vectors.

Linear Algebra : Study of linear systems, which are set of eqns. involving linear form of variables.

Scalars

Vectors

Matrices

Tensors.

Why linear algebra in ML?

→ Generalizing concepts in higher dimensions.

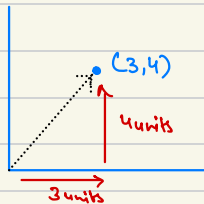
(Humans have tough times dealing with 4D & above dimensions data)

→ Data Representation : Wide variety of data.

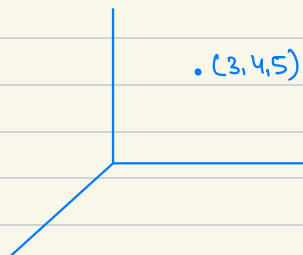
Like tabular, text, image, video etc.

What are vectors?

2D Co-ordinates.



$a = [3, 4]$
↑ ↖
x-component y-component
vector of dimension $\rightarrow 2$



$a = [3, 4, 5]$
in 3D space.
vector of dimension $\rightarrow 3$

n-dimensions

$$a = [x_1, x_2, \dots, x_n]$$

vector of dimension $\rightarrow n$

Vectors
 $\downarrow$

iris = 150 flower species

	input vector				target vector
	SL	SW	PL	PW	species
datapoints $\rightarrow$	5.2	2.1	2.2	9.1	setosa
	3.1	4.2	7.0	6.5	versicolour

I/P $\rightarrow$ feature vector = $[5.2, 2.1, 2.2, 9.1]$ $\rightarrow$ [model] $\rightarrow$ O/P = Setosa.

Titanic Dataset.

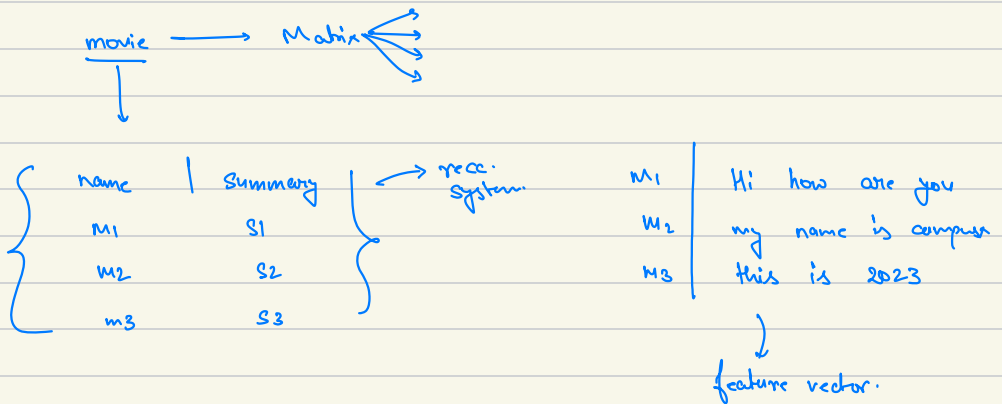
age	fare	sex	Pclass	Embarked
17	52	M	1	S

feature vector $\rightarrow [17, 52, 0, 1, 2]$

encoding $\begin{matrix} M & F \\ 1 & 0 \end{matrix}$

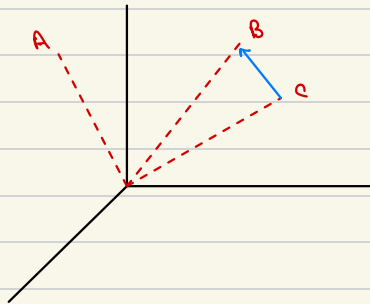
$\begin{matrix} P & C & S \\ 0 & 1 & 2 \end{matrix}$

Vectors example in ML :



3 feature vector $\downarrow$ 100 days

		Hi	How	are	you	my	name	is	computer	this	2023
$\rightarrow$	1	1	1	1	0	0	0	0	0	0	0
$\rightarrow$	0	0	0	0	1	1	1	1	0	0	0
$\rightarrow$	0	0	0	0	0	0	1	0	1	1	1



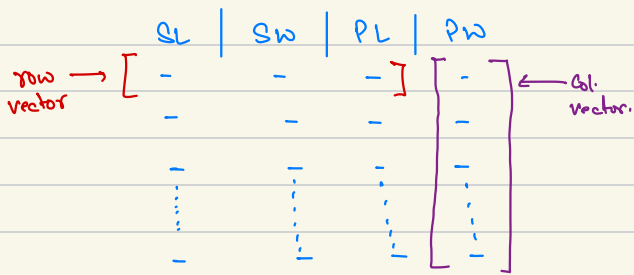
Row and Column Vector. :

$a = [x_1, x_2, \dots, x_n]$

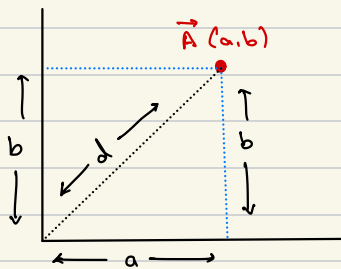
$\begin{array}{c} \nearrow \text{row vector} \\ \text{row} \uparrow \text{col.} \\ 1 \times n \end{array}$

$b = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$

$\begin{array}{c} \nearrow \text{col. vector} \\ n \times 1 \\ \text{rows} \leftarrow \text{col.} \end{array}$



Distance from origin $\|A\|$:



$$d = \sqrt{a^2 + b^2}$$

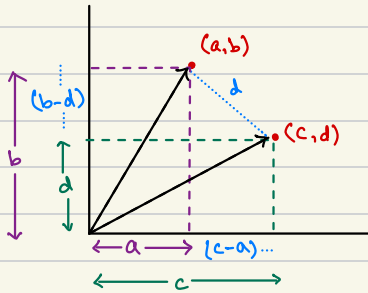
$$A(a, b, c) \quad d = \sqrt{a^2 + b^2 + c^2}$$

$$nd \quad d = \sqrt{x_1^2 + x_2^2 + \dots + x_n^2}$$

Also known as magnitude of vector $\|A\|$.

Python : $d = \text{np.linalg.norm}(A)$

Euclidean Distance :



$$\text{dist} = \sqrt{(c-a)^2 + (b-d)^2}$$

$$\text{dist} = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}$$

Python :

```
A = np.array([1, 2, 3])  
B = np.array([4, 5, 6])
```

$$\text{diff} = A - B$$

$$\text{dist} = \text{np.linalg.norm}(\text{diff})$$

Uses : KNN uses euclidean distance internally.

Scalar Addition / Subtraction (Shifting) :

Scalar
Addition / Subtraction

Scalar
Multiplication / Division.

$$v = [v_1, v_2, v_3, \dots, v_n]$$

$$v = [v_1 + n, v_2 + n, \dots, v_n + n]$$

$$v = [v_1 - n, v_2 - n, \dots, v_n - n]$$

} Shifting

$$v = [v_1, v_2, v_3, \dots, v_n]$$

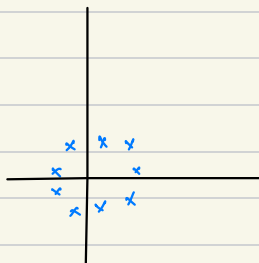
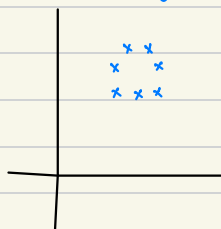
Use : Mean Centering

eg.) 50 vectors 2D system.

	x_1	x_2	x_1 shifted	x_2 shifted
v_1	3	4	$3 - 5 = -2$	-2
v_2	5	6	0	0
v_3	7	8	2	2
			$\Sigma = 0$	0

$$\mu_{x_1} = 5$$

$$\mu_{x_2} = 6$$



mean centered.

Dot Product :

$$A = [a_1, a_2, \dots, a_n]$$

$$B = [b_1, b_2, \dots, b_n]$$

Dot product (scalar)

$$a_1 b_1 + a_2 b_2 + \dots + a_n b_n$$

Can be treated as matrix multi.

$$\begin{matrix} [a_1, a_2, \dots, a_n] \\ \text{1xn} \end{matrix} \begin{matrix} \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{bmatrix} \\ \text{nx1} \end{matrix}$$

= (x1) matrix.

vectors $\rightarrow$ by default we take col. vectors.

$$\begin{matrix} \begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{bmatrix} \\ a^T \end{matrix} \quad \begin{matrix} \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{bmatrix} \\ b \end{matrix}$$

$$A \cdot B = A^T B$$

Rules : Commutative $[A \cdot B = B \cdot A]$

Distributive $[A \cdot (B+C) = A \cdot B + A \cdot C]$

np.dot(A, B)

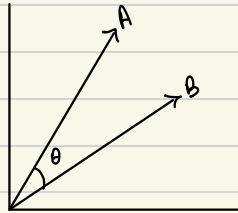
or $A @ B$

Use : Compute similarity b/w 2 vectors.

Calculating projections

Matrix multiplication

Angles b/w 2 vectors. :



$$A \cdot B = \|A\| \|B\| \cos \theta$$

Distance from origin

A, B both are non zero

1.) $A \cdot B = 0$

$$\rightarrow \|A\| \|B\| \cos \theta = 0$$

$\downarrow$
 90°

$A \perp B$ (orthogonal)

A & B are dissimilar

$$A = [a_1, a_2, \dots, a_n]$$

$$B = [b_1, b_2, \dots, b_n]$$

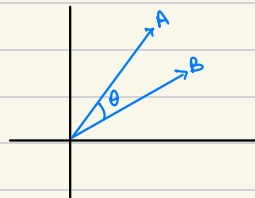
$$\theta = \cos^{-1} \frac{A \cdot B}{\|A\| \|B\|}$$

Cosine Similarity :

$$\cos \theta = \frac{A \cdot B}{\|A\| \|B\|}$$

Cos of angle b/w A & B.

In ML $\rightarrow$ used for similarity measure.



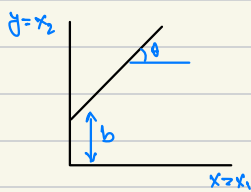
$$\cos \theta = [-1 \text{ to } 1]$$

if val = 1, $\theta = 0^\circ$
 $\therefore$ Similar.

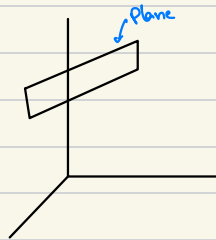
if $\theta = 180^\circ$, polar opposite

if $\theta = 90^\circ$, $\perp$ orthogonal
 $\hookrightarrow$ dissimilar.

Equation of a Hyperplane :



2D $\rightarrow$ line



3D - Plane

4D, 5D, ... nD $\rightarrow$ Hyperplane.

line: $y = mx + c$
 $\downarrow$ general form.

$$ax + by + c = 0$$

$$y = -\frac{a}{b}x - \frac{c}{b}$$

$$m = -\frac{a}{b}, b = -\frac{c}{b}$$

$$\text{Plane: } w_1x_1 + w_2x_2 + w_3x_3 + w_0 = 0$$

$$\text{Hyperplane: } w_1x_1 + w_2x_2 + w_3x_3 + \dots + w_nx_n + w_0 = 0$$

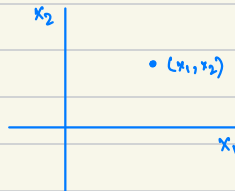
Hyperplane represented by π

π in form of vectors.

$$ax + by + c = 0$$

$$ax_1 + bx_2 + c = 0$$

$$w_1x_1 + w_2x_2 + w_0 = 0$$



$$w_1x_1 + w_2x_2 = 0$$

$$[w_1, w_2] = w \cdot x$$

$$w \cdot x = w_1x_1 + w_2x_2$$

$$\begin{matrix} \swarrow \\ \bullet \end{matrix} (x_1, x_2, x_3) \quad w \cdot x = w_1x_1 + w_2x_2 + w_3x_3$$

$$w = [w_1, w_2, \dots, w_n]$$

$$x = [x_1, x_2, \dots, x_n]$$

$$w_1 x_1 + w_2 x_2 + w_3 x_3 + \dots + w_n x_n + w_0 = 0$$

↓

$$w \cdot x + w_0 = 0$$

$$\begin{matrix} [w_1, w_2, w_3 \dots w_n] & \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ \vdots \\ x_n \end{bmatrix} \\ 1 \times n & n \times 1 \end{matrix}$$

$$w^T \cdot x + w_0 = 0$$

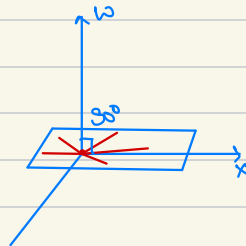
↓ $w_0 = 0$ [passing from origin]

$$w^T \cdot x = 0$$

$$w^T \cdot x = 0$$

$$w^T \cdot x = w \cdot x = \|w\| \|x\| \cos \theta = 0$$

$$\theta = 90^\circ$$



Linear Algebra | Matrices (computation)

Matrix

$$M = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix} (3 \times 3)$$

↳ elements.

Order of matrix : $(n\text{-rows} \times n\text{-cols})$
(Shape)

Matrix is a mathematical tool.

Uses :

- Linear system of equation.
- Linear Transformation.
- Eigen value and Eigenvectors.
- Graph Theory.
- Markov Chain
- Computer Graphics.
- Control theory.
- Optimization.

Normal Matrix

$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$$

Row Matrix

$$\begin{bmatrix} 1 & 2 & 3 & 4 \end{bmatrix}$$

Column Matrix

$$\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$$

Square Matrix

$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}_{3 \times 3}$$

→ Diagonal of matrix

Diagonal Matrix

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix}$$

Scalar Matrix →

Identity Matrix →

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

→ Diagonal of Matrix have same value.

Zero Matrix

$$\begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 0 & 0 \end{bmatrix}$$

Matrix A = Matrix B [equal matrix]

1.) Order should be same

2.) $A_{ij} = B_{ij}$

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$$

$$B = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$$

$$A = B$$

Scalar Operations :

$k \rightarrow$ Scalar

$A \rightarrow$ Matrix

$k = 2$

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$$

$$k+A \Rightarrow \begin{bmatrix} 3 & 4 \\ 5 & 6 \end{bmatrix}$$

$$kA \Rightarrow \begin{bmatrix} 2 & 4 \\ 6 & 8 \end{bmatrix}$$

if $k = -1$

$$kA \Rightarrow \begin{bmatrix} -2 & -4 \\ -6 & -8 \end{bmatrix} \Rightarrow -A.$$

Rule : $A, B, k.$

$$k(A+B) = kA + kB.$$

Matrix Addition & Subtraction :

Criteria : A, B : Order should be same.

$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}_{2 \times 2} + \begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix}_{2 \times 2} \Rightarrow \begin{bmatrix} 6 & 8 \\ 10 & 12 \end{bmatrix}_{2 \times 2}$$

Subtraction : $A - B \Rightarrow A + (-1) B$

$\uparrow$ $\uparrow$
Scalar scalar
Add. multi

Rule : $A + B = B + A$ (Commutative)

$(A + B) + C = A + (B + C)$ (Associative)

1. Additive Identity.

$$A + X = A$$

$$\begin{bmatrix} 2 & 3 \\ 4 & 5 \end{bmatrix} + X = \begin{bmatrix} 2 & 3 \\ 4 & 5 \end{bmatrix}$$

$$X = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

Acts as additive identity.

2. Additive Inverse.

$$A - X = 0$$

$$\begin{bmatrix} 2 & 3 \\ 4 & 5 \end{bmatrix} - X = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

$$-A = \begin{bmatrix} -2 & -3 \\ -4 & -5 \end{bmatrix}$$

Acts as additive inverse.

Matrix Multiplication :

dot product.

$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}_{2 \times 2} \times \begin{bmatrix} 4 & 5 \\ 6 & 7 \end{bmatrix}_{2 \times 2} = \begin{bmatrix} 1 \times 4 + 2 \times 6 & 1 \times 5 + 2 \times 7 \\ 3 \times 4 + 4 \times 6 & 3 \times 5 + 4 \times 7 \end{bmatrix}_{2 \times 2}$$
$$= \begin{bmatrix} 16 & 19 \\ 36 & 43 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}_{2 \times 3} \times \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}_{3 \times 3} = \begin{bmatrix} 1+8+21 & 2+16+24 & 3+12+27 \\ 4+20+42 & 8+25+48 & 12+30+54 \end{bmatrix}$$
$$= \begin{bmatrix} 30 & 36 & 42 \\ 66 & 81 & 96 \end{bmatrix}$$

Rule : $A \cdot B \neq B \cdot A$ (Not Commutative)
 $(A \cdot B) \cdot C = A \cdot (B \cdot C)$ (Associative)
 $A(B+C) = AB + AC$ (Distributive)

Multiplicative Identity

$$A \cdot I = A.$$

$$I = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

(identity matrix)

$$\text{i.e. } \boxed{A \cdot I = A}$$

Transpose of Matrix :

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$$

$$A^T = \begin{bmatrix} 1 & 4 & 7 \\ 2 & 5 & 8 \\ 3 & 6 & 9 \end{bmatrix}$$

Transpose : $m \times n = n \times m$

Symmetric Matrix : $A = A^T$

eg. $A = \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}$ $A^T = \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}$

Skew-Symmetric Matrix : $A^T = -A$

Rules :

$$\begin{aligned} (A^T)^T &= A \\ (A+B)^T &= A^T + B^T \\ (AB)^T &= B^T A^T \end{aligned}$$

Determinants :

$$A x = B$$

$$\begin{bmatrix} 1 & 2 \\ 6 & 5 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 4 \\ 11 \end{bmatrix}$$

$$x = B A^{-1}$$

Only Sq. matrix is inverted.

$$A^{-1} = \frac{1}{\det(A)} \text{Adj}(A)$$

non-square $\rightarrow$ pseudo inverse.

Determinant is a scalar value, computed from a square matrix.

$$A = [1] \Rightarrow \det(A) = [1]$$

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \Rightarrow \det(A) = A = \begin{vmatrix} 1 & 2 \\ 3 & 4 \end{vmatrix} = 1 \times 4 - 6 = -2.$$

How to determine if inverse of a matrix is possible?

1. Square Matrix.
2. $\det(A) \neq 0$

$\det(A) = 0$ called singular matrix.

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$$

$$\det(A) = \Delta = \begin{vmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{vmatrix}$$

$$\det(A) = 1(45-48) - 2(36-42) + 3(32-35)$$

$$\det(A) = -3 + 12 - 9$$

$$\det(A) = 0$$

Minor :

$$A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$$

$$\det(A) \rightarrow \begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix}$$

$$M_{11} = a_{22}$$

$$M_{12} = a_{21}$$

$$M_{21} = a_{12}$$

$$M_{22} = a_{11}$$

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$$

$$M_{11} = \begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix}$$

$$= a_{23} \times a_{32} - a_{22} \times a_{33}$$

Co-factor :

$$A_{ij} = (-1)^{i+j} \times M_{ij} \quad \xrightarrow{\text{Minor}}$$

$$A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$$

$$A_{11} = (-1)^{1+1} = (-1)^{1+1} \times M_{11} = (-1)^2 \times a_{22}$$

$$A_{12} = (-1)^3 \times a_{21}$$

Adjoint :

Also called adjugate / classical adjoint.

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$$

find cofactor of all elements & transpose it.

$$A^T = \begin{bmatrix} A_{11} & A_{21} & A_{31} \\ A_{12} & A_{22} & A_{32} \\ A_{13} & A_{23} & A_{33} \end{bmatrix}$$

Inverse :

$$A^{-1} = \frac{1}{\det(A)} \operatorname{adj}(A)$$

Solving a system of linear equation :

$$\begin{cases} x + y = 5 \\ 4x + 3y = 15 \end{cases} \quad [x, y] = ?$$

$$\begin{matrix} \begin{bmatrix} 1 & 1 \\ 4 & 3 \end{bmatrix} & \begin{bmatrix} x \\ y \end{bmatrix} & = & \begin{bmatrix} 5 \\ 15 \end{bmatrix} \\ A & x & & B \end{matrix}$$

$$A \cdot x = B$$

$$A^{-1} A \cdot x = A^{-1} B$$

$$I x = A^{-1} B$$

$$x = A^{-1} B$$

$$A = \begin{bmatrix} 1 & 1 \\ 4 & 3 \end{bmatrix}$$

$$A^{-1} = \frac{1}{\det(A)} \operatorname{Adj}(A)$$

$$\det(A) = 3 - 4 = -1$$

$$A = \begin{bmatrix} 1 & 1 \\ 4 & 3 \end{bmatrix}$$

Finding cofactor

$$A = \begin{bmatrix} 3 & -4 \\ -1 & 1 \end{bmatrix}$$

$$A_{11} = (-1)^{1+1} \cdot 3 = 3$$

$$A_{12} = (-1)^{1+2} \cdot 4 = -4$$

$$A_{21} = (-1)^{2+1} \cdot 1 = -1$$

$$A_{22} = (-1)^{2+2} \cdot 1 = 1$$

Adjoint

$$A^T = \begin{bmatrix} 3 & -1 \\ -4 & 1 \end{bmatrix}$$

Inverse.

$$A^{-1} = \frac{1}{\det(A)} \text{Adj}(A)$$

$$A^{-1} = \frac{1}{-1} \begin{bmatrix} 3 & -1 \\ -4 & 1 \end{bmatrix}$$

$$A^{-1} = \begin{bmatrix} -3 & 1 \\ 4 & -1 \end{bmatrix}$$

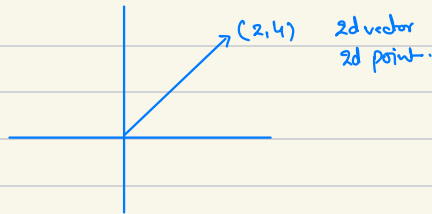
$$\text{for } x, y \Rightarrow A^{-1} \cdot B \Rightarrow \begin{bmatrix} -3 & 1 \\ 4 & -1 \end{bmatrix} \begin{bmatrix} 5 \\ 15 \end{bmatrix} \Rightarrow \begin{bmatrix} -15+15 \\ 20-15 \end{bmatrix} \Rightarrow \begin{bmatrix} 0 \\ 5 \end{bmatrix}$$

$$x = 0, y = 5.$$

Linear Algebra | Metrics (intuition)

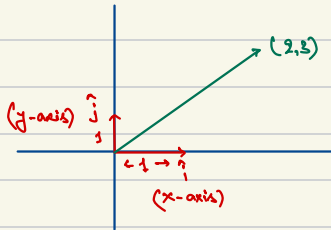
Vectors : Set of no's.

$$\begin{array}{ccc} [2, 3, 4] & , & [2, 4] \\ \downarrow & & \downarrow \\ \text{dimension} = 3d & & \text{2d vector.} \end{array}$$



Basis Vectors :

2d co-ordinate system.



Distance from origin = 1.

$$2i + 3j \quad (\text{Linear Combination.})$$

Linear Transformation :

Transformation : $f(x)$

input $\rightarrow f(x) \rightarrow$ output.

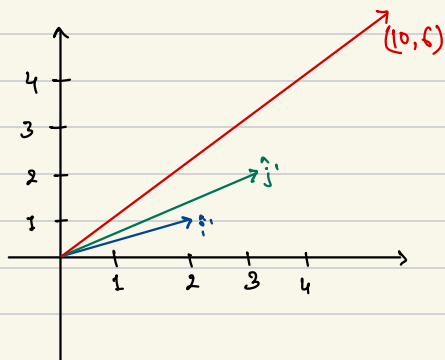
vector as input $\rightarrow$ Transformation $\rightarrow$ another vector
Matrix.

vector $\times$ matrix $\rightarrow$ another matrix.

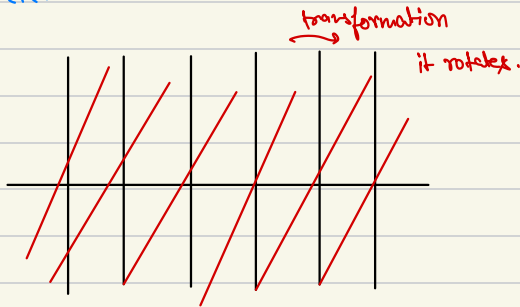
$\uparrow$ it works as transformation.

$$\begin{bmatrix} \hat{i} & \hat{j} \\ 2 & 3 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} 2 \\ 2 \end{bmatrix} \rightarrow \begin{bmatrix} 10 \\ 6 \end{bmatrix}$$

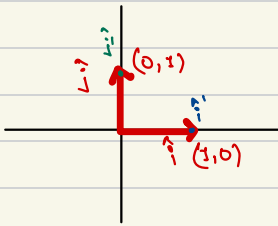
Transformation Matrix A Transformed.



Grid after transformation of vectors. generally rotates or transformed to stretch, expand etc.



Why there is no transformation in identity matrix?

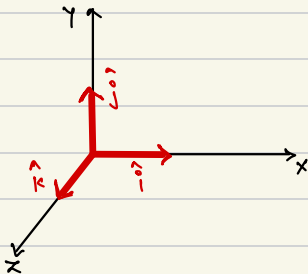


$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

Same co-ordinate points, thus, no transformation.
therefore, it's called identity matrix.

Vector transformation in 3d. :



$$A \times \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix} \longrightarrow \text{Transformed.}$$

Transformation Matrix.

Why called Linear transformation?

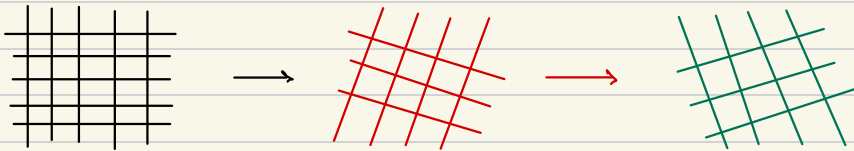
1. Origin doesn't change.
2. After applying transformation, grid lines are parallel, they will never intersect.

Matrix Multiplication as Composition. :

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \quad B = \begin{bmatrix} 2 & 3 \\ 4 & 8 \end{bmatrix}$$

Grid $\longrightarrow$ Performed Transformation B $\xrightarrow{\text{results to}}$ New Grid structure
(Let's call it Red grid)

Red Grid $\longrightarrow$ Performed Transformation A $\xrightarrow{\text{results to}}$ New transformation
(Let's call it Green grid)



Test of Commutative law :

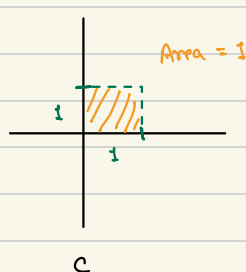
$$A \cdot B \neq B \cdot A$$

$C \rightarrow A \rightarrow C' \rightarrow B \rightarrow C''$
 $C \rightarrow B \rightarrow C' \rightarrow A \rightarrow C''$ } Both are different
(Visualise as matrix multiplication)

Determinant :

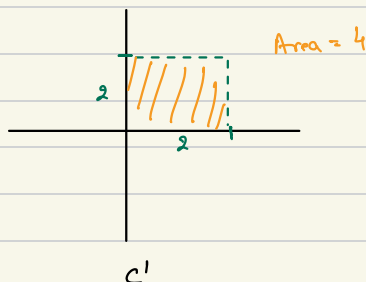
$$A = \begin{vmatrix} 2 & 3 \\ 4 & 5 \end{vmatrix} = 5 \times 2 - 4 \times 3 = 10 - 12 = -2.$$

↑
What does it represent?
→ Volume scaling factor for linear transformation.



$$\begin{bmatrix} \hat{i} & \hat{j} \\ 2 & 0 \\ 0 & 2 \end{bmatrix}$$

Transformation.



$$\det(A) = \frac{\text{Area}'}{\text{Area}} = \frac{4}{1} = 4$$

$\det(A)$ tells ratio of Areas after transformation.

Scaling factor of area → Determinant.

Why we cannot find inverse if $\det(M) = 0$?

→ Because all the points are flattened to a line in a 2d space.

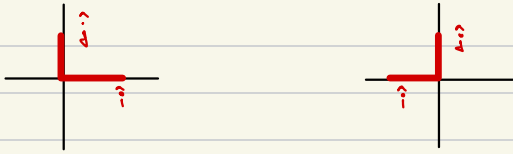
Thus, leading area = 0.

Basically transforming 2d space $\xrightarrow{}$ 1d transformation.

Why $\det(M)$ is possible only for sq. matrix?

→ Because dimensions remain same. (e.g. 2d → 2d only)

What does it mean to have a 've' det.?



→ i, j flips in 've' determinant.

Inverse :

$$A \rightarrow A^{-1}$$

$$A^{-1} A \cdot x = A^{-1} B$$

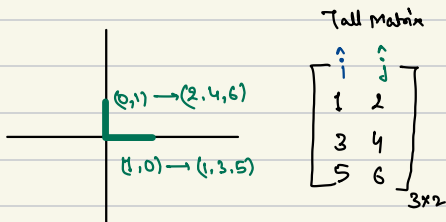
$$I \cdot x = A^{-1} B$$

$$x = A^{-1} B.$$

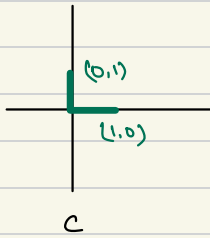
$$\begin{array}{ccccc} A & \cdot & x & = & B \\ \uparrow & & \uparrow & & \uparrow \\ \text{Transformation} & & \text{input vector} & & \text{New Vector Output.} \end{array}$$

$A^{-1} \rightarrow$ Reverting the transformation to get back to A.

Transformations for non-square matrix :



Transforming 2d space $\rightarrow$ 3d space



Wide Matrix

$$\begin{bmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$$

Transforming 3d space $\rightarrow$ 2d space.

$\therefore$ We cannot find determinant, as we can't find area from one dimension to another dimension.

Why only sq. M have inverse?

$\rightarrow$ Because a non-sq. matrix will be either tall or wide.

Hence, change of dimension, so cannot find the area.

We lose the information of one dimension or we cannot map the info. of one dimension in new vector.

Why inverse is possible for non-singular matrices only?

$$\downarrow$$

$$\det(A) \neq 0$$

$\rightarrow$ Change of dimension from 2d/3d $\rightarrow$ 1d

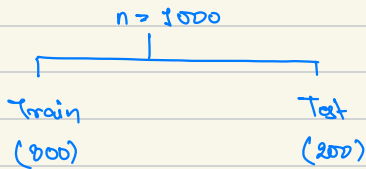
As all the points lie in a line, hence area = 0.

We lose the information.

Classification Matrices Pt.1 | Accuracy and Confusion Matrix | Type 1 and 2 Errors

$n = 1000$

Cgpa	I ₂	Lpa
-	-	-
-	-	-
-	-	-



Training models in Logistic Regression and Decision Tree.



Carry out predictions

Accuracy Label	I.R Prediction	D.T. Prediction
1	1 ✓	1 ✓
0	1 ✗	1 ✗
0	0 ✓	0 ✓
0	0 ✓	0 ✓
1	1 ✓	1 ✓
1	1 ✓	1 ✓
0	1 ✗	0 ✓
0	0 ✓	0 ✓
0	0 ✓	0 ✓
1	1 ✓	1 ✓

↑ prediction ✓ or ✗.

We will check the accuracy of both the models for each student.

$$\text{accuracy} = \frac{\text{No. of Correct Prediction}}{\text{Total predictions}}$$

$$\text{accuracy (L.R)} = \frac{8}{10} = 0.8 \approx 80\%$$

$$\text{accuracy (D.T)} = \frac{9}{10} = 0.9 \approx 90\%$$

D.T is performing well than compare to L.R.

Accuracy of Multi-classification Problem :

Actual Label	L.R. Predict	D.T Predict.
0	0	0
0	0	0
0	0	0
2	2	2
0	0	0
2	2	2
0	0	0
2	2	2
1	1	1
1	1	1

$$\text{accuracy-score} = \frac{10}{10} \rightarrow 1 \approx 100\%$$

How much accuracy is good?

→ There is no fixed answer, it depends on the problem we are solving.

eg) For swiggy → 90% accuracy also works.

for Cancer Detection → 99% accuracy models also fails, as we cannot afford to loose 1%.

Thus, accuracy is subjective.

The Problem with Accuracy.

Accuracy score gives a number (eg. 90%), which mean 10% are still the mistakes.

It does not tells about what type of mistakes are being made.

Confusion Metrics :

		Prediction	
		1	0
Actual	1	TP	FN
	0	FP	TN

		1	0	
Act.	1	26	6	Model says - No Actually - Yes
	0	0	29	Correct Predictions : Model says → No Actual → No.

Model - No Predicted
Actual - Yes.

$$\text{Accuracy} = \frac{\text{TP} + \text{T.N}}{\text{TP} + \text{T.N} + \text{FP} + \text{FN}}$$

Type 1 Error : Total no. of false positive (FP) is called Type 1 Error.

Type 2 Error : Total no. of false negative (FN) is called Type 2 Error.

		1	0	
Actual	1		FN	Type 2 Error.
	0	FP		Type 1 Error

Predicted.

Classification Matrix for Multiclassification Problem :

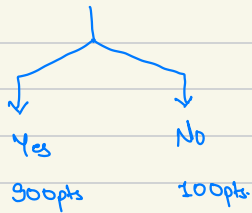
		Predicted		
		0	1	2
Actual	0	7	0	5
	1	2	21	6
	2	3	1	13

True Predictions.

$$\text{Accuracy} = \frac{\text{True Predictions}}{\text{Totals}}$$

When is accuracy misleading ?

→ Imbalanced Dataset : When the ratio of the 2 classes is imbalanced.



		1	0
Actual	1	0	1
	0	0	99999

Prediction

$$\text{Accuracy} \rightarrow \frac{99999}{99999+1} \approx 99.999\%$$

↑
Due to imbalanced dataset

Part 2 | Precision, recall, F1 score

Accuracy can be misleading sometimes.
Therefore we need precision, recall and F1 score.

Precision :

Model A : Spam Classifier.

Actual \ Prediction	1	0
1	100 TP	170 FN
0	30 FP	700 TN

$$\text{Accuracy} = \frac{800}{1000} = 80\%$$

$$\text{Precision}_A = \frac{100}{100+30}$$

Model B : Spam Classifier.

Actual \ Prediction	1	0
1	100 TP	130 FN
0	10 FP	700 TN

$$\text{Accuracy} = \frac{800}{1000} = 80\%$$

$$\text{Precision}_B = \frac{100}{100+10}$$

$$F \cdot P_A > F \cdot P_B$$

Type 1 Error
Precision.

$$F \cdot N_A < F \cdot N_B$$

Type 2 Error.

$\therefore$ We will deploy Model B.

$$\text{Precision} = \frac{TP}{TP + FP}$$

Recall :

Model A : Chest Cancer

		1	0
Actual	1	1000 TP	200 FN
	0	900 FP	9000 TN
		Predicted	

Accuracy : 90%

$$\text{Recall}_A = \frac{1000}{1200}$$

$R_A > R_B$.

Model B : Chest Cancer.

		1	0
Actual	1	100 TP	500 FN
	0	500 FP	9000 TN
		Predicted	

Accuracy : 90%

$$\text{Recall}_B = \frac{100}{1500}$$

F.P → Do not have cancer, model predicted have Cancer.

F.N → Had Cancer, Model predicted Not Cancer.

$$F.N_A < F.N_B$$



We will deploy Model A.

Recall : What proportion of actual positives is correctly classified?

$$\text{Recall} = \frac{TP}{TP + FN}$$

F1 Score :

Sometimes it's difficult to decide among which error is more needed.

To solve it we use F1 score.

It's a combination of precision & recall (Harmonic Mean)

$$\boxed{F1 \text{ score} = \frac{2PR}{P+R}}$$

Harmonic mean generally leans towards low value.

eg) $P=0$, $R=100$

$$F1 \text{ score} = \frac{P+R}{2} \text{ (Arithmetic Mean)} = \frac{100}{2} = 50$$

$$F1 \text{ score} = \frac{2PR}{P+R} \text{ (Harmonic Mean)} = \frac{2 \cdot 0 \cdot 100}{0+100} = 0$$

eg. 2)

$$P=R=80$$
$$F1 = \frac{2 \cdot 80 \cdot 80}{160} = 80$$

$$P=60 \quad R=100$$
$$F1 = \frac{2 \cdot 60 \cdot 100}{160} = 75$$

Model B is penalized due to low P.

Hence, Harmonic mean tends to lean toward lower side.
(Here, Precision)

Multiclass Precision and Recall :

Behind the scene, we calc. the scores for all even in binary classification.
But we take value of 1 (positive) or 0 (Negative) depending we are choosing precision or recall.

		Predicted			
		Dog	Cat	Rabbit	Total
Actual	Dog	25	5	10	40
	Cat	0	30	4	34
	Rabbit	4	10	20	34
	Total	29	45	34	108

$$\text{Precision}_{\text{Dog}} = \frac{25}{29} = 0.86$$

$$\text{Precision}_{\text{cat}} = \frac{30}{45} = 0.66$$

$$\text{Precision}_{\text{Rabbit}} = \frac{20}{34} = 0.58$$

Combined Precision :

$$\text{Macro Precision} = \frac{0.86 + 0.66 + 0.58}{3} = 0.70$$

$$\text{Weighted Precision} = 0.86 \times \frac{40}{108} + 0.66 \times \frac{34}{108} + 0.58 \times \frac{34}{108} \Rightarrow 0.71$$

		Predicted			
		Dog	Cat	Rabbit	Total
Actual	Dog	25	5	10	40
	Cat	0	30	4	34
	Rabbit	4	10	20	34
	Total	29	45	34	108

$$\text{Recall}_{\text{Dog}} = \frac{25}{40} = 0.62$$

$$\text{Recall}_{\text{cat}} = \frac{30}{34} = 0.88$$

$$\text{Recall}_{\text{Rabbit}} = \frac{20}{34} = 0.58$$

Combined Recall :

$$\text{Macro Recall} = \frac{0.62 + 0.88 + 0.58}{3}$$

$$\text{Weighted Recall} = 0.62 \times \frac{40}{108} + 0.88 \times \frac{34}{108} + 0.58 \times \frac{34}{108}$$

		Predicted				
		Dog	Cat	Rabbit	Total	Recall
Actual	Dog	25	5	10	40	0.62
	Cat	0	30	4	34	0.88
	Rabbit	4	10	20	34	0.58
	Total	29	45	34	108	
Precision		0.86	0.66	0.58		

$$F1 \text{ score}_{\text{dog}} = \frac{2 \cdot P_d \cdot R_d}{P_d + R_d}$$

$$F1 \text{ score}_{\text{cat}} = \frac{2 \cdot P_c \cdot R_c}{P_c + R_c}$$

$$F1 \text{ score}_{\text{Rabbit}} = \frac{2 \cdot P_r \cdot R_r}{P_r + R_r}$$

Combined F1 score:

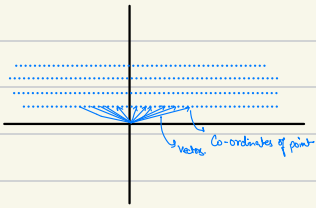
Macro F1 score

Weighted F1 score

Eigen Values and Eigen Vectors

Linear Transformation, Eigen Vectors and Eigen Values. :

Matrices are Linear transformations.



Transformation of Matrix.
(Span & Magnitude changes)

Eigen vectors $\rightarrow$ Special vectors, when transformation is applied, these vectors does not changes the direction (span).

Eigen values $\rightarrow$ The factor at which the values of eigen vectors changes is called Eigen Values.

Rules: $A \vec{v} = \lambda \vec{v}$

$A \rightarrow$ Matrix

$\vec{v} \rightarrow$ Eigen vector.

$\lambda \rightarrow$ Eigen value

Linear Transformations are like movement, the eigen vector remains at its position and acts as the axis of rotation.

Calculating Eigen Vectors :

$$\begin{bmatrix} 2 & 3 \\ 1 & 1 \end{bmatrix}$$

$$A\vec{x} = \lambda\vec{x}$$

$$I \cdot A\vec{x} = \lambda I\vec{x}$$

$$A\vec{x} - \lambda I\vec{x} = 0$$

$$(A - \lambda I)\vec{x} = 0$$

$(A - \lambda I) \rightarrow$ Makes it zero vector.

↑
Non-invertible matrix
i.e. $\det = 0$

$$\det(A - \lambda I) = 0$$

$$\det \left(\begin{bmatrix} 2 & 3 \\ 1 & 1 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \right) = 0$$

$$\det \left(\begin{bmatrix} 2-\lambda & 3 \\ 1 & 1-\lambda \end{bmatrix} \right) = 0$$

$$(2-\lambda)(1-\lambda) + 3 = 0$$

$$2 - 2\lambda - \lambda + \lambda^2 + 3 = 0$$

$$\lambda^2 - 3\lambda + 5 = 0$$

$$\lambda^2 - 3\lambda + 5 = 0$$

$$x = \frac{b \pm \sqrt{b^2 - 4ac}}{2a}$$

We get the complex roots, i.e. an imaginary eigen vector exists in the system.

$$A = \begin{bmatrix} 2 & 3 \\ 0 & 1 \end{bmatrix}$$

$$\det \left(\begin{bmatrix} 2 & 3 \\ 0 & 1 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \right) = 0$$

$$\det \left(\begin{bmatrix} 2-\lambda & 3 \\ 0 & 1-\lambda \end{bmatrix} \right) = 0$$

$$(2-\lambda)(1-\lambda) - 0 = 0$$

$$(2-\lambda)(1-\lambda) = 0$$

$$\lambda = 1, 2$$

Eigen values(λ) = 1, 2.

$$\text{Eigen vectors} \rightarrow (A - \lambda I) \vec{v} = 0$$

$$(A - \lambda I) \vec{v} = 0$$

$$\left(\begin{bmatrix} 2 & 3 \\ 0 & 1 \end{bmatrix} - 2 \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \right) \cdot \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = 0$$

$$\left(\begin{bmatrix} 2 & 3 \\ 0 & 1 \end{bmatrix} - \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix} \right) \cdot \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = 0$$

$$\begin{bmatrix} 0 & 3 \\ 0 & -1 \end{bmatrix} \cdot \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = 0$$

$$3x_2 = 0$$

$$x_2 = 0$$

Here $x_1 = k$ (any value)

$\vec{v} = (1, 0), (2, 0), (3, 0), \dots, (k, 0)$
(Eigen vectors)

for $\lambda = 3$

$$\left(\begin{bmatrix} 2 & 3 \\ 0 & 1 \end{bmatrix} - \begin{bmatrix} 3 & 0 \\ 0 & 3 \end{bmatrix} \right) \cdot \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = 0$$

$$\begin{bmatrix} -1 & 3 \\ 0 & 0 \end{bmatrix} \cdot \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = 0$$

$$x_1 + 3x_2 = 0$$

$$x_1 = -3x_2$$

So, if $x_2 = 1 \rightarrow x_1 = -3$ $[-3, 1]$ eigen vector.

$$A\vec{v} = 0$$

it means $\lambda\vec{v} = 0$, so the space is squishing down from higher dimension to lower dimension (e.g. 3d $\rightarrow$ 2d) or (e.g. 2d $\rightarrow$ 1d)

Properties of EV :

1. Sum of eigen values = sum of elements in a diagonal.

2. Product of eigen values = determinant of that matrix (M)

3. Eigen vectors corresponding to different eigenvalues are orthogonal :

If a matrix A is symmetric (i.e. $A = A^T$), the EV they are orthogonal (perpendicular) to each other, they are not co-related. ↗ perpendicular.

4. EVs of Identity Matrix (I) : $EVs(I) = 1$

5. EVs of a scalar multiple : $B = cA$ λ will be $c\lambda$

6. EVs of diagonal matrix : λ = same as diagonal.

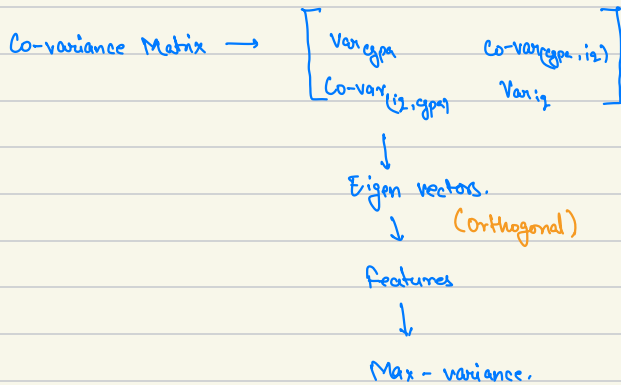
$$\begin{bmatrix} 3 & 0 \\ 0 & 4 \end{bmatrix} \quad \lambda = 3, \lambda = 4$$

7. EVs of Transpose Matrix : $\lambda(A) = \lambda(A^T)$

Eigen Vectors in PCA :

cgpa | iq | lpa

Feature Extraction.



Eigen Decomposition + PCA Variants

Diagonal Matrix :

$$A = \begin{bmatrix} a & 0 \\ 0 & b \end{bmatrix} \quad A = \begin{bmatrix} a & 0 & 0 \\ 0 & b & 0 \\ 0 & 0 & c \end{bmatrix}$$

Diagonal Matrix
stores.

- Power $\rightarrow A^{100} \rightarrow \begin{bmatrix} a^{100} & 0 \\ 0 & b^{100} \end{bmatrix}$

- Eigen value $\rightarrow \lambda(A) = a, b$

- Multiplication by a vector $\rightarrow \begin{bmatrix} x \\ y \end{bmatrix}$ component scale by 'a' factor
component scale by 'b' factor.

- Matrix Multiplication $\rightarrow \begin{bmatrix} a & 0 \\ 0 & b \end{bmatrix} \begin{bmatrix} c & 0 \\ 0 & d \end{bmatrix} \rightarrow \begin{bmatrix} ac & 0 \\ 0 & bd \end{bmatrix}$

Orthogonal Matrix :

It's a sq. matrix, whose columns & rows are orthogonal unit vectors (i.e., orthonormal vectors), meaning that they are all of unit length and are at right angles to each other.

$$\begin{bmatrix} \vec{v}_1 & \vec{v}_2 \\ a & b \\ c & d \end{bmatrix} \rightarrow \text{orthogonal}$$

- Properties:
 $\vec{v}_1 \perp \vec{v}_2$
 $\angle 90^\circ$
i.e. $ab + cd = 0$

$$\sqrt{a^2 + c^2} = 1$$

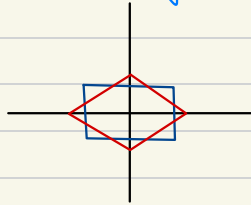
$$\sqrt{b^2 + d^2} = 1$$

$I \in$ Orthogonal Matrix.

Inverse of Orthogonal Matrix = Transpose of Orthogonal matrix.

$$A^{-1} = A^T$$

Perfect rotation, no scaling and shearing.



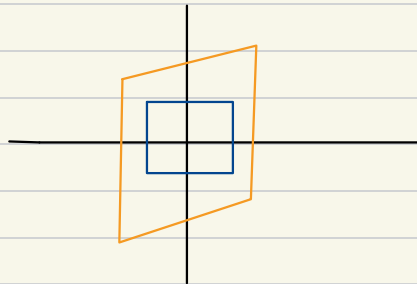
Symmetric Matrix :

Square Matrix

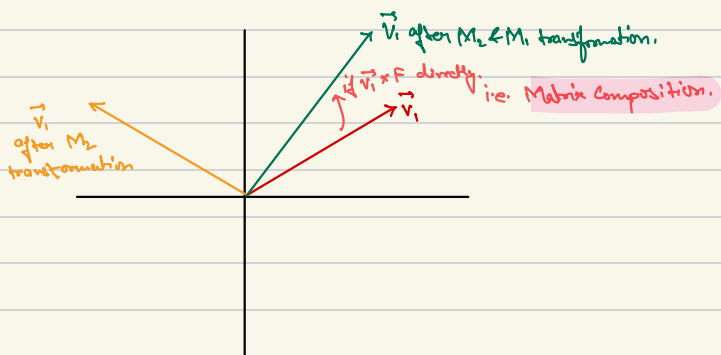
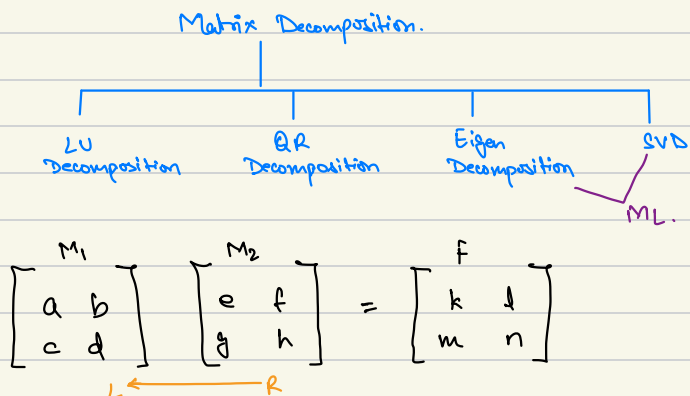
$$A = A^T$$

eg) $A = \begin{bmatrix} a & c \\ c & b \end{bmatrix} \quad A^T = \begin{bmatrix} a & c \\ c & b \end{bmatrix} \Rightarrow \text{Symmetric matrix.}$

- Real Eigen Values.
- Orthogonal Eigenvectors.



Matrix composition :-



vice versa of Matrix composition is Matrix Decomposition.

Eigen Decomposition :-

The eigen decomposition of a matrix A is given by the equation:

$$A = V \Lambda V^{-1}$$

- V is a matrix whose columns are eigenvectors of A .
- Λ is a diagonal matrix whose entries are eigenvalues of A .
- V^{-1} is inverse of V .

Assuming :

- Square Matrix.
- Diagonalizability : for a $n \times n$ matrix it should have a linearly independent eigen vectors.
↳ different directions eigen vectors. 

$$\text{eg:)} \quad \begin{bmatrix} a & b \\ c & d \end{bmatrix} \quad \lambda_1 \quad \lambda_2 \quad \begin{bmatrix} x_1 \\ y_1 \end{bmatrix} \quad \begin{bmatrix} x_2 \\ y_2 \end{bmatrix}$$

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} x_1 & x_2 \\ y_1 & y_2 \end{bmatrix} \begin{bmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{bmatrix} \begin{bmatrix} x_1 & x_2 \\ y_1 & y_2 \end{bmatrix}^{-1}$$

Eigen vector : $A \vec{v} = \lambda \vec{v}$

$$A \vec{v}_1 = \lambda_1 \vec{v}_1$$

$$A \vec{v}_2 = \lambda_2 \vec{v}_2$$

$$A V = V \Lambda$$

$$V = [\vec{v}_1 \quad \vec{v}_2]$$

$$\vec{v}_1 = \begin{bmatrix} x_1 \\ y_1 \end{bmatrix} \quad \vec{v}_2 = \begin{bmatrix} x_2 \\ y_2 \end{bmatrix}$$

$$\Lambda = \begin{bmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{bmatrix}$$

$$A \vec{v}_1 = \lambda_1 \vec{v}_1$$

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix}_{2 \times 2} \begin{bmatrix} x_1 \\ y_1 \end{bmatrix}_{2 \times 1} = \lambda_1 \begin{bmatrix} x_1 \\ y_1 \end{bmatrix}$$

$$\begin{bmatrix} ax_1 + by_1 \\ cx_1 + dy_1 \end{bmatrix} = \lambda_1 \begin{bmatrix} x_1 \\ y_1 \end{bmatrix}$$

$$ax_1 + by_1 = \lambda_1 x_1 \quad \text{--- (1)}$$

$$cx_1 + dy_1 = \lambda_1 y_1 \quad \text{--- (2)}$$

$$A \vec{v}_2 = \lambda_2 \vec{v}_2$$

$$ax_2 + by_2 = \lambda_2 x_2 \quad \text{--- (3)}$$

$$cx_2 + dy_2 = \lambda_2 y_2 \quad \text{--- (4)}$$

$$AV = \Lambda V$$

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} x_1 & x_2 \\ y_1 & y_2 \end{bmatrix} = \begin{bmatrix} x_1 & x_2 \\ y_1 & y_2 \end{bmatrix} \begin{bmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{bmatrix}$$

$$\begin{bmatrix} ax_1 + bx_2 & ax_2 + by_2 \\ cx_1 + dy_1 & cx_2 + dy_2 \end{bmatrix} = \begin{bmatrix} \lambda_1 x_1 & \lambda_2 x_2 \\ \lambda_1 y_1 & \lambda_2 y_2 \end{bmatrix}$$

$$ax_1 + bx_2 = \lambda_1 x_1$$

$$ax_2 + by_2 = \lambda_2 x_2$$

$$cx_1 + dy_1 = \lambda_1 y_1$$

$$cx_2 + dy_2 = \lambda_2 y_2$$

$$A \leq \lambda I$$

$$\longrightarrow Av = v\Lambda \longrightarrow$$

$$A = V \Lambda V^{-1}$$

inverse V

eigen
vectors

diagonal
eigen
values

Eigen Decomposition of Symmetric Matrix :-

$$A = V \Lambda V^{-1}$$

spectral decomposition.

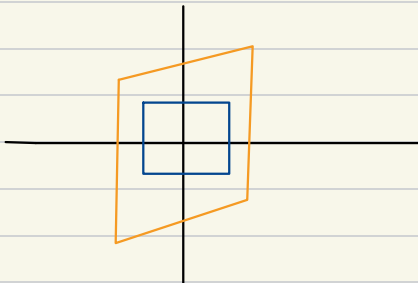
$v \rightarrow$ Orthogonal

$\Lambda \rightarrow$ Diagonal matrix.

Symmetric matrix $\xrightarrow{\text{Decompose}}$ Orthogonal matrix and diagonal matrix.

$$A = V \Lambda V^{-1}$$

V → rotation
 Λ → scaling
 V^{-1} → rotation



Advantages of Eigen Decomposition :-

M2 (PCA)

Physics

Signal Processing.

Quantum Physics.

$$A_{n \times n} = A^{1000}$$

$$\hookrightarrow A = V \Lambda V^{-1}$$

$$A^{100} = V \Lambda V^{-1} \cdot V \Lambda V^{-1} \cdot V \Lambda V^{-1} \dots$$

$$A^{100} = V \Lambda^{100} V^{-1}$$

$$A^{100} = V \begin{bmatrix} \lambda_1^{100} & 0 \\ 0 & \lambda_2^{100} \end{bmatrix} V^{-1}$$

